Audit Regulation, Auditor Industry Expertise and Effects on Audit Pricing

Imen BEN SLIMENE

University of Haute Alsace-CREGO (EA 7317) University of Bourgogne-Franche Comté

Abstract

The diversity of regulations that govern statutory audit in European countries provides the opportunity to analyse how audit regulation affects audit fees. This study is the first to analyse empirically the joint effect of audit regulation and auditor expertise on audit fees for the institutional setting of 14 European countries by using a sample of 4,293 European firms for the period from 2003 to 2011. I analyse four attributes of audit regulation, namely terms for annual renewal of the mandate, nature of the auditors' liability, whether the state requires joint audits and whether it imposes restrictions on the provision of non-audit services. The main results show that using an industry specialist auditor results in higher audit fees. Moreover, three regulatory attributes—auditor liability, annual renewal of the mandate and joint audit—play a significant role in determining audit fees. Fees are higher when the auditor's mandate may be renewed or terminated each year, and when the auditor's liability is tort. Mandatory joint audit has also a positive impact on audit fees.

Keywords: Audit fees, Audit regulation, Tenure and audit rotation, Auditors' liability, Auditor Industry Expertise.

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1. INTRODUCTION

Recognizing the crucial impact of regulation on the determination of audit fees, several studies examining audit fees around the world have focused on the impact of the Sarbanes-Oxley Act (SOX) on audit fees in the USA. Nevertheless, these studies have only compared the level of audit fees in the pre- and post-SOX periods (Ghosh and Pawlewicz 2009; Evans and Schwartz 2014) and in the international context (André et al. 2016); they have not considered the impact of national regulation of the audit market and, more importantly, the single effect of each attribute of such regulation on audit fees.

To my knowledge, the study of André et al. (2016) is the first to examine the effect of regulation of audit fees in a European context. Besides the degree of investor protection, it appears that other institutional factors play a key role in explaining audit fees. André et al. (2016) reference the most widely used legal attributes (Wingate 1997; La Porta et al. 1998; Djankov et al. 2008) to classify France, Italy and UK. They consider that audit fees in UK should be higher than in France and Italy, the audit fees being higher as investor protection is strong. Their results show that audit fees are higher in France than in the UK, suggesting that other factors explain the differences in fee levels between countries. Their study suffers from two limitations I propose to raise here: a) it covers only three European countries, and b) the authors compare the audit fees paid in each country regardless of the specific national regulations.

This study contributes to the literature by considering the diversity of statutory audit regulations that prevails in the various member states of the European Union (EU). More specifically, I focus on the role of the national audit regulations and the quality of auditors in explaining the differences in audit fees between European countries. To do so, I take into consideration four attributes of audit regulation that strongly differentiate the regulations in force in the fifteen EU member states under study, namely auditor tenure, auditor liability, joint audit and provisions of non-audit services. Based on a cross-country study from 2003 to 2011, the main results show that, in addition to the classical determinants of audit fees (auditor reputation, investor protection, firm size, leverage, audit risk, etc.), three variables in state regulation—nature of auditor liability, ease of annual renewal of the mandate and requirement of joint audit—play a significant role in determining audit fees. Fees are higher when the auditor's mandate may be renewed or not renewed at the end of each year and when the auditor's liability is tort. Mandatory joint audit has also a positive impact on audit fees.

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The remainder of the paper is organized as follows. The second section presents the literature related to the impact of different attributes of the regulations that influence audit fees. The third section describes my research design. The fourth section analyses and discusses the main results. To ensure the robustness of these results, the fifth section presents several additional tests. The last section offers concluding remarks and identifies the main contributions of the research.

2. BACKGROUND AND RESEARCH QUESTION DEVELOPMENT

2.1. Audit regulation and audit fees

The gradual adoption of International Standards on Auditing (ISA), the recommendations issued in 2000, 2002 and 2003 by the European Commission to ensure audit quality in Europe, the adoption of the new Eighth Directive in 2006 and the publication of the Green Paper on Audit Policy in October 2010, which led to the adoption of the Directive on Statutory Audit in April 2014, aim to improve and harmonize audit services in the EU. However, the Statutory Audit Directive leaves the member states some areas of discretion regarding the implementation of certain elements, and national characteristics persist regarding audit regulation. These discrepancies come from the EU composition of heterogeneous countries with influential national authorities.

The national audit markets are quite different with regard to the regulatory and legislative environment of financial reporting and auditing (Quick 2008; Le Vourc'h and Morand 2011). A review of the regulatory environment for statutory audit in the EU shows that national audit regulations differ based on four attributes: the minimum duration of auditor tenure, the legal basis for auditors' liability, the constraints on the joint supply of audit and non-audit services and the requirement of a joint audit.

Civil liability of statutory auditors

Given the risk of litigation, auditors can enhance their effort or charge an insurance premium to cover the financial consequences of any disagreements (Simon and Francis 1988; Simunic and Stein 1996; Bedard and Johnstone 2004; Badertscher et al. 2014). Arbitration between the costs associated with these actions and potential sanctions for litigation sheds an interesting light on many issues, including audit fees and audit quality (Dye 1993; Willekens et al. 1996; Ye and Simunic 2013).

Previous studies have shown a positive relationship between litigation risk and audit pricing. However, each study addresses this issue from a different angle (Minutti-Meza 2014). Lennox

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and Li (2012) analyse the impact of a reduction of auditor liability on audit quality and audit fees, i.e. the transition from the unlimited auditor liability regime to limited liability (LLP) in the UK. Their results show that audit fees and market share of audit firms did not change significantly following the adoption of limited liability (LLP). Lennox and Li (2012) explain the stability of fees by a lack of competition in the UK audit market. He et al. (2017) analyse the impact of legal liability audit quality and audit fees. (Since 2010, Chinese audit firms were required to transform from a structure of limited liability company (LLC) to limited liability partnership (LLP), which removes the cap on the liability exposure of negligent auditors.) Their results show that auditors are more likely to charge a premium in audit fees after audit firms reorganize as LLPs, which suggests that exerting unlimited legal liability on negligent auditors raises both audit quality and audit fees.

In Europe, the debate on auditor liability is at the heart of many reflections. While some countries limit the liability of auditors (Austria, Belgium, Germany, Greece and Slovenia), others restrict the liability limit to what is stipulated by contract (UK) or know no limitation of liability. In cases where the liability regime is not limited, auditing firms may tend to refuse to work for some high-risk firms. In a highly litigious country, the costs charged to firms for a statutory audit may be higher than elsewhere.

The London Economics and Ewert study (2006) aimed at analysing the economic impact of auditors' liability regimes. It investigated whether the differences between countries in civil liability regimes have an impact on audit fees. Given the limited data available, the researchers conducted a questionnaire survey of audit firms. They asked if the audit fees varied depending on the degree of the audit engagement risk and whether this risk was fully integrated into audit fees. The majority of respondents (83% of firms in general and 70% of 'Big' audit firms) indicated that the level of audit fees varies according to the mission risk.

Based on the influence outlined here, the first hypothesis regarding the effects of auditor liability regimes on audit fees is as follows:

H 1.1. The audit fees are higher when the liability of the auditor is strong.

In this study, I test this hypothesis empirically by considering the auditors' liability regimes in the different European countries. Consistent with Defond (2012), I expect that audit fees can also be affected by many other factors, including regulations concerning auditor tenure and joint audit. These two dimensions are the subject of the following sections.

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Auditor tenure regulation

The literature presents a set of circumstances that may lead firms to change their auditors: a change in leadership, auditor-auditee conflict, economics difficulties or a regulatory constraint. The Anglo-Saxon countries (i.e. the United States and the United Kingdom) allow firms to change their auditors every year. However, some continental European countries limit change of auditors by imposing mandatory multi-year audit mandates (three years in Belgium, six years in France). In this study, I treat this relationship from a regulatory point of view. In other words, I examine the impact of the possible effect of annual renewal of the auditor's mandate on audit fees. Will firms that have the option of changing their auditors each year because of regulations pay a premium or benefit from lower fees?

A renewable long-term audit mandate allows auditors to achieve economies of scale and, as a result, lower audit fees. However, a long-term audit mandate is likely to strengthen the relationship between the auditor and the audited firm (with the risk of collusion between them) and thus increase audit fees. The auditor's change can lead to higher fees because of high start-up costs but also because of the practice of 'opinion shopping'.[1] However, changing the auditor may also allow the customer to benefit from a reduction of fees due to the strong competition in the audit market and the practice of 'low-balling'. These developments suggest that the length of the auditor-auditee relationship affects the amount of the fees charged by the auditor. However, it is difficult to predict the direction of the relationship. Based on these developments, the second hypothesis is as follows:

H 1.2- The possibility of annual renewal of the auditor's mandate leads to higher audit fees.

Mandatory joint audit

Although a joint audit is not mandated by the new European Directive on statutory audit, this measure is strongly recommended. France is the only country that enforces a joint audit. In Denmark, it was mandatory until 2005.

The main reason for opposing a joint audit is the potential increase in costs related to the appointment of two auditors (André et al. 2016). According to André et al. (2016), higher fees result from costs of coordination and the joint liability of the two auditors. They are in fact

[1]In countries where renewable audit mandates are annual, firms can take advantage of the latitude available to them to adopt such behavior of 'opinion shopping'. However, widespread economic consequences of this practice have not been proven empirically (Lennox et al. 2000, 2002; Choi et al. 2010, 2015).

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jointly responsible for the audit, so each auditor assumes the risk of a failure by the other. Thus, the joint auditors may request a risk premium (André et al. 2016). This premium is usually requested by the Big when they work with non-Big (deep pockets hypothesis: Gonthier-Besacier and Schatt 2007).

Several empirical studies have analysed the impact of a joint audit on audit fees. Most of them have analysed the impact of the mandatory joint audit on audit fees especially in France and Denmark (Gonthier-Besacier and Schatt 2007; Audousset-Coulier 2015; Lesage et al. 2017).

André et al. (2016) show that the French firms subject to mandatory joint audit pay higher audit fees than the Italian and British firms. They explain the higher audit fees by three main reasons. Firstly, the presence of two auditors increases costs of coordination. Second, the two auditors are legally jointly liable, meaning that one auditor assumes the risk of the failure of the other. Third, mandatory long-term audit tenure in France (six years) limits the opportunity for managers to renegotiate the audit fees.

I analyse the impact of joint audit on audit fees in 14 European countries. Referring to this literature, the hypothesis regarding the effects of mandatory joint audit on audit fees is as follows:

H 1.3- The audit fees are higher when a joint audit is mandatory.

Provision of non-audit services

The provision of non-audit services is a crucial issue given the developing relationship between auditor and audited firm. The provision of non-audit services is often presented as the main threat to auditor independence (Bartlett 1993; Canning and Gwilliam 1999; Callaghan et al. 2009). It appears further that the fees charged by audit firms for non-audit services have continued to grow (Beattie et al. 2003).

Empirical research on the relationship between audit fees and the provision of non-audit services presents two divergent views. First is a positive association between audit fees and non-audit fees (Hay 2013). According to Francis et al. (2006), companies that use the same auditor for audit and non-audit services pay higher fees (Simunic 1984; Francis et al. 2005). The auditor devotes more effort when he or she jointly provides external audit and non-audit services (Davis et al. 1993).

Second, the joint provision of audit and non-audit services allows auditors a better understanding of the client's business (knowledge spillover). It could facilitate the auditor's work and reduce the cost of the audit (Simunic 1984; Craswell and Francis 1999; Whisenant et al. 2003) and therefore the audit fees. In the European context, several recent studies validate this finding (Antle et al.

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2006; Clatworthy and Peel 2007 in the United Kingdom; DeFuentes et al. 2010 in Spain; Raffournier and Schatt 2013 in Switzerland).

Raffournier and Schatt (2013) explain the positive relationship between audit and non-audit fees by the fact that most of the research has been conducted in the United States before the adoption of SOX in 2002. Francis et al. (2005) show that mandatory disclosure of audit fees in the annual reports of US firms (from 2001) resulted in a change in the determination of audit fees. Furthermore, as the fees for audit and non-audit work are simultaneously determined (DeFond et al. 2002; Whisenant et al. 2003), it is possible that the prohibition on jointly providing audit and non-audit services affects the determination of audit fees (Fung et al. 2012).

In this context and in view of the heterogeneity of the EU Member States' rules concerning the joint provision of audit and non-audit services, I postulate that:

H 1.4- The joint provision of audit and non-audit services positively affects the audit fees.

2.2. Auditor industry expertise and audit fees

Studies on the determination of audit fees have shown that large international audit firms receive consistently higher fees (Choi et al. 2008; Choi et al. 2009; Ghosh and Pawlewicz 2009; Francis and Yu 2009; Fung et al. 2012). This positive relationship between Big firms and auditors' remuneration is also confirmed by studies across countries analysing the determinants of audit fees (Fargher et al. 2001; Choi et al. 2008). Choi et al. (2008) show the existence of an audit fee premium for Big firms, a premium more pronounced in countries where legal systems are less stringent.

In the European context, research addressing the relationship between auditor size and audit fees has yielded mixed results. In France, Gonthier-Besacier and Schatt (2007) and André et al. (2016) prove a positive relationship with a higher premium when the auditor is Big. Similarly in Germany, a Krauss et al. (2015) study shows that Big receive higher fees than other auditors. In contrast, Antle et al. (2006) in the UK and Cameran (2005) in Italy don't find a significant relationship between auditor size and audit fees. There results confirm Knechel and Willikens's (2006) results in Belgium.

Industry specialization is one potential reason for differences in audit quality and audit fees among the Big. These firms started promoting their expertise in certain industries and hence used this as a way to differentiate their product from competing Big auditors. If this strategy is successful, fee premiums should be observed for specialist auditors. The meta-analysis of Hay et

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al. (2006) and its extension by Hay (2013) strongly support the hypothesis that large audit firms can demand higher fees because of their reputation and industry specialization.

Most of the previous research implies a positive relationship between auditor industry specialization and audit fees; however, these results depend on the definition of auditor industry specialization (Hay 2013). The effect of this specialization on audit fees shows mixed results (Fung et al. 2012). Such a relationship can be positive. That reflects the effect of auditor reputation. Several studies find that firms are willing to pay a premium to secure the services of an auditor specialist (Palmrose 1986; Craswell et al. 1995; Ferguson et al. 2003; Francis et al. 2005; Carson 2009; Johnstone al. 2014; etc.). This relationship may instead be negative due to economies of scale in the provision of audit services (Ferguson and Stokes 2002; Mayhew and Wilkins 2003; Cahan et al. 2008).

In the particular context of IPOs, Mayhew and Wilkins (2003) analysed the impact of industry specialization of audit firms on audit fees. They show that specialization allows audit firms to achieve competitive advantages in terms of cost and quality of service. Nevertheless, the impact of these benefits on audit fees depends on the differentiation of the firm relative to its competitors. Only audit firms that have significantly higher market share than their competitors get premium industry audit fees. Based on these findings, I advance the following hypothesis:

H.2.1 - There is a positive relationship between auditor industry expertise and audit fees.

3. RESEARCH DESIGN

3.1. Variables measures

Measures of audit regulation

In order to examine the relation between audit market regulations and audit fees in Europe, I focus on the information on a wide set of regulations contained in Table 1. I have collected information on various audit regulations from national laws. I have relied on studies by Le Vourc'h and Morand (2011) published by the European Commission, Quick et al. (2008), Ratzinger-Sackel et al. (2012) and Hess (2014) and on the responses to the Assessment of the Regulatory and Standard-Setting Framework and Standards Development Questionnaire published by IFAC as part of the Agency Member Compliance Program. I have verified and updated the information directly with the national laws and regulations of each country in the sample. More precisely, I use country-specific dummy variables for each year between 2003 and 2011 which indicate whether regulation has been effective.

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The nature of the liability of the auditor

The legal forms of auditors' liability differ widely across countries. Liability rules can be embedded in contract law, tort law or both. An audit failure that causes a purely financial loss to a third party can be regarded as a violation of contractual duties between the auditor and the third party, or just as a tort. In contract law, pure economic losses are generally compensated.

Under tort law, compensations are strongly restricted or even excluded. To assess how differences in regimes of auditor liability may affect audit fees, I create a variable (*Liability)* that refers to the statutory auditors' legal liability.

Liability is coded 1 if the liability of auditors is based on tort law and 0 otherwise.

According to hypothesis H.1.1, audit fees are measured by the logarithm of *LnFees* fees charged and are being positively related to *Liability*.

The minimal duration of the audit mandate

To analyse the effects of audit regulation relating to audit tenure, I have classified the EU countries in my sample according to the minimal length of the audit mandate. Table 1 synthesizes the state of European national regulations on the matter.

R_Mandat is a dummy variable coded 1 if the mandate of the external auditors may be renewed on a yearly basis. Consequently, *R_Mandat* equals 0 if the minimum length of the audit mandate is mandatorily more than 1 year.

Under hypothesis H.1.2, audit fees (*LnFees*) are expected to be positively related to the ability to change auditors every year.

Mandatory joint audit

In Europe only France requires a joint audit. In Denmark, joint audit was mandatory until 2005. I have created a variable *Joint_A* that takes the value 1 if the joint audit is mandatory, 0 otherwise. According to the hypothesis H.1.3, I expect a positive relationship between audit fees and the requirement to audit jointly.

The provision of non-audit services

I consider three levels of regulatory constraints regarding the provision of non-audit services. Non-audit services may be fully allowed, restricted or fully banned. To determine how differences in audit regulation concerning the joint provision of audit and non-audit services

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impact audit fees, I create a dummy variable (*NAS*) that takes the value 1 if the joint provision of audit and non-audit services is constrained (limited or prohibited), 0 otherwise.

According to the hypothesis H.1.4, I expect a positive relationship between audit fees and authorization to charge for non-audit services.

The variables characterizing the attributes of audit regulation in the 14 EU member states are summarized in Table 1.

[Insert Table 1]

Measures of auditor industry expertise

The most widely used hypothesis is that large audit firms can demand higher fees because of their reputation and industry specialization (Hay et al. 2006). Thus, I use a dummy variable *BIG4* that equals 1 if the firm is audited by at least one Big auditor, 0 otherwise. I expect to see a positive correlation between *LnFees* and *BIG4*.

Several studies assume that audit firms gain expertise and specialization through experience in an industry. Following Craswell et al. (1995), Ferguson et al (2003) and Mayhew and Wilkins (2003), I retain the market share of an audit firm in a given industry as a proxy for auditor industry specialization. I employ two levels of industry specialization: a national level and European level. Thus, the *SP-E* variable equals 1 if the market share of an audit firm, in terms of total assets in a given industry as defined by the Global Industry Classification Standard (GICS) at the EU level, is greater than or equal to 10%. *SP-C* takes the value 1 if the audit firm's portion of the market, in terms of total assets of firms audited by industry and by country, is greater than or equal to 25%.

Control variables

The audit literature suggests three determinants of audit fees: client size, complexity and risk (Hay et al. 2006; Hay 2013). The majority of empirical research shows a positive relationship between these determinants and audit fees (Simunic 1980; Choi et al. 2008; Francis et al. 2005; Fung et al. 2012). I measure the size of the firm *LnTA* as the natural logarithm of total assets. Based on prior literature, audit complexity and client risk are represented in this study by the following measures: the proportion of inventories and receivables in total assets (*INREC*). Working for clients with operating losses (*Loss*) is expected to be riskier, suggesting higher audit fees (e.g., Larcker and Richardson 2004; Francis et al. 2005; Hay et al. 2006). Leverage (*LEV*) is also a proxy of clients' insolvency risk (e.g., Craswell et al. 1995; Craswell and Francis 1999; Larker and Richardson 2004; Francis et al. 2005; Hay et al. 2006).The client firm's industry can influence the

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litigation risk and at the same time the level of effort provided by the auditor and therefore the audit fees (Causholli et al. 2010). Thus, I include a dummy variable, *SICRISK*, that equals 1 if the firm belongs to an industry with a significant risk of litigation, 0 otherwise.

Two explanatory variables, 'first-year audit engagement' and 'busy season', were selected to define the engagement attributes in this study. The variable *P-Year* equals1 if it is the first year of the auditor's engagement. I expect a negative relationship between this variable and audit fees. The fee reduction may be due to the practice of 'low-balling' (DeAngelo 1981; Ghosh and Lustgarten 2006; Ghosh and Pawlewicz 2009). Auditors are known to have a 'busy season' that corresponds mainly with 31 December. It is hypothesized that an audit conducted during the busy season may be more costly and, consequently, there will be a positive relationship with audit fees. December is a dummy variable equal to 1 if the firm closes its accounts on 31 December, 0 otherwise.

I include macroeconomic variables as control variables that influence demand of audit services and therefore the audit fees (Choi et al. 2008). Previous literature shows that the transition to IFRS is associated with an increase in audit fees. *IFRS* is a dummy variable coded 1 if the year of the audit is after the year of transition to IFRS by the country where the firm is domiciled. It is 0 otherwise.

André et al. (2016) and Choi et al. (2008) include the gross domestic product (*GDP*) and the foreign direct investment of a country (*FDI*) as institutional factors. They show a positive relationship between GDP and audit fees. The variables *GDP* and *FDI* are yearly metrics taken from *World Bank*.

3.2. Regression model

To test the impact of both auditor quality (i.e. audit firms' size and expertise) and audit regulation on fees charged to European firms, I use the following model:

$$LnFees = \alpha_0 + \alpha_1 (BIG4 / SP\text{-}E / SP\text{-}C) + \Sigma\beta_i REG_i + \Sigma\chi_j Control_j + EF + \varepsilon$$

LnFees captures the total audit fees. The variables BIG4, SP-E and SP-C are proxies of auditor quality (i.e. audit firm size and auditors' industry specialization). The REG_i variables characterize audit regulations affecting auditor tenure, civil liability of auditors, mandatory joint audit and the joint provision of audit and non-audit services. The $Control_j$ variables represent the 'usual' determinants of the audit fees presented by previous studies (Hay et al. 2006; Hay 2013). The fixed effects (*EF*) aim to control both for general trends common to all firms in a given year and for time-invariant heterogeneity across industries that may not be captured by the control

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variables. I also cluster standard errors by year and industry in order to minimize the lack of independence of residuals (error terms) between year and industry (Petersen 2009). ε is an error term.

3.3. Sample Selection

The initial sample included all listed firms in the 15 countries that were members of the EU before its expansion to the countries of Eastern Europe: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, Spain, Sweden and the United Kingdom. Data were collected from OSIRIS and Thomson Worldscope databases. I focused on a nine-year period, from 2003 to 2011. I obtained an initial sample consisting of 38,637 observations (5,960 firms over 9 years).

I believe that the choice of this period helps strengthen the quality of the research and contribute to a better understanding of audit fees. This period is particularly interesting for several reasons. First, the year 2003 followed a number of landmark and historic events for auditors in terms of implementing rules and codes of corporate governance at the international, European and national levels. Secondly, the study period covers two important years for the EU, 2005 and 2006. In order to harmonize the practices of European companies, the EU implemented the International Financial Reporting Standards (IFRS) in 2005 and adopted the Eighth European Directive on Statutory Audits of Annual Accounts and Consolidated Accounts in 2006 (Directive 2006/43/EC of the European Parliament and of the Council). Then came the publication of the Green Paper on Audit Policy in October 2010, which led to the adoption in April 2014 of the Directive on Statutory Audit that was implemented on June 2016.[2] This directive concerns the annual accounts and the consolidated accounts and aims to strengthen and harmonize the legal control in EU member countries. Indeed, despite the harmonization attempts made by the European Commission, national audit environments within Europe vary greatly in terms of independence rules and auditor liability. Therefore, restrictions imposed by national audit environments can be expected to influence audit quality and consequently audit fees among member countries.

Indeed the study period also covers the years following the financial scandals at the beginning of this century and the global financial crisis of 2007–2010. In Europe, these financial failures have raised questions about external auditors and audit committees and their performance in the

[2] I chose to conclude with the year 2011 because the objective of this research is to analyse the impact of the diversity of auditing regulations between EU member countries and therefore before the adoption of the new directive.

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financial markets. In the context of recent auditing regulations, much of the focus has been on the organization of external audits, the audit committee and the separation of audit fees and non-audit fees to preserve the independence and the audit quality. The question of whether these measures were sufficient to meet the public demand is still a subject of concern for market regulators.

The final sample consisted of 11,515 observations (firm-year). I excluded all finance companies such as banks, financial institutions, and insurance companies due to their specific accounting rules. I eliminated from the sample firms with missing data. I also eliminated firms with abnormal data values, such as zero or negative total assets or sales.

Steps of the constitution of the sample

Number of companies listed in the 15 EU countries surveyed	**5,960**
-Banks (SIC codes 6000–6799)	1,631
-Firms listed in Luxemburg	36
Number of companies	**4,293**
Firm-year observations from 2003 to 2011	38,637
Missing data	27,122
- Missing data on audit fees	18,249
- Other missing data	8,873
Number of observations	**11,515**

Table 2 reports descriptive statistics for the remaining variables. I present average values of these variables by country. The mean of audit fees paid by firms in the sample is € 9.7 million. The higher fees paid by Netherlands firms average € 106 million, followed by French companies (€ 28 million) and Italian firms (€ 24 million). Greek companies are the smallest, with total assets of 489 million euros on average. Thirty-six point one percent of European firms in the sample are audited by industry specialist auditors. It appears that 56.5% of the sampled firms are audited by a Big Four, but this proportion varies strongly from one country to the other. Sixty-nine percent of the sample companies apply IFRS. Finally, 74.42% of companies close their financial years on 31 December.

[Insert Table 2]

To avoid a possible collinearity problem, I produce the matrix of correlation coefficients of the variables used in the regressions (Table 4). There is a strong positive correlation between the log of audit fees (*LnFees*) and the size of audited firm (*LnTA*).

If the correlation coefficient exceeds 78%, it can be concluded that company size plays a fundamental role in the billing of audit fees in Europe. As expected, there is a positive correlation

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between Big Four audit firm and auditor industry specialization (*SP-E* and *SP-C*), as the majority of specialist auditors are Big. The size of the firm (*LnTA*), the size of the audit firm (*BIG4*) and the auditor industry specialization variables (*SP-E* and *SP-C*) show a strong positive correlation.

Large firms are generally audited by a Big, often a specialist, and thus pay higher fees. The variables of the duration of mandate (*R_Mandat*) and mandatory joint audit (*Joint-A*) are highly correlated. This correlation is found in the specificity of the French context: France is the only country that requires firms to have two auditors, and the minimum duration of the mandate is also relatively long (six years). Therefore, to avoid any bias multicollinearity, I do not include these variables simultaneously in the regression.

[Insert Table 3]

4. Analysis of results

This study examines the relationship between audit regulation and audit fees. I have chosen in the first instance to introduce the attributes of audit regulation one by one in order to analyse the individual effect of each attribute. In the second instance, I simultaneously introduce all the regulation attributes. The results are reported in Table 4.[3]

[Insert Table4]

The first three columns of Table 4 show the results of the impact of the possibility of annual renewal of the auditor's mandate (*R_Mandat*) on audit fees. The coefficients of *R_Mandat* are positive and statistically significant in all models. The possibility of renegotiating the terms annually leads to higher audit fees. The start-up costs related to weak knowledge of the audited firm and the lack of familiarity with its accounting procedures result in higher audit fees (Cameran 2005). The risks of the inability to transfer mentioned costs over several years make this argument stronger.

The results relating to the civil auditor's liability (*Liability*) are reported by columns 4 to 6 of Table 4. In conformity with my hypothesis, the audit fees are higher when the auditor's liability is tort. This result is also consistent with previous literature showing that audit fees increase with the auditor's litigation risk. Auditors increase audit efforts as well as risk premiums in response to higher litigation risk (Choi et al. 2008; Ghosh and Pawlewicz 2009; André et al. 2016). In both cases, the fees are positively impacted.

[3]I have successively introduced audit firm size (*BIG4*), auditor specialization in European level (*SP-E*) and local level (*SP-C*), due to high correlation between these variables.

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The mandatory joint audit (*Joint-A*) had a positive and significant effect on audit fees. This positive relationship is consistent with the results of most of the empirical studies in the French context. André et al. (2016) explain higher audit fees by the fact that the joint audit potentially needs high organization and coordination costs, which are reflected in the audit fees. Restrictions on the provision of non-audit services (*NAS*) lead to lower total fees. This result is the logical consequence of the restriction of the provision of non-audit services in the majority of European countries.

The last three columns in Table 4 summarize the results of the joint effect of audit regulatory attributes on audit fees. Included simultaneously, all the key variables retain their expected signs. Tort auditor liability (*Liability*) and the possibility of annual renewal of the auditor's mandate (*R_Mandat*) are associated with higher audit fees. These results confirm the results previously presented. Auditor attributes (i.e. Big Four and auditor industry specialization at the European level *SP-E* or national level *SP-C*) impact the audit fees positively. These results confirm previous research finding that a fee premium is charged by Big Four for their services (Choi et al. 2008; Jaggi and Low 2011; André et al. 2016) and by specialists (Carson 2009; Fung et al. 2012; Johnstone et al. 2014).

Control variables that capture 'traditional' determinants of audit fees retain the expected signs and significance for all models, thus confirming the results of earlier literature. The size of the audited firm (*LnTA)* is positively and significantly associated with audit fees. The sums of accounts receivable and inventory (*INREC*) have positive effects on audit fees. But the significance of this variable is not systematic. This result can be explained by the less complex character of these assets in auditing. These elements, which traditionally had a higher risk, are easier to audit after the adoption of IFRS. LEV has a positive and significant effect on audit fees. This result corroborates the results of Choi et al. (2008), Jaggi and Low (2011) and André et al. (2016). The positive relationship between *IFRS* and audit fees confirms the hypothesis that the period of transition to *IFRS* is associated with an increase in audit fees (Kim et al. 2012). Consistent with the hypothesis of low-balling practice, *P-Year* is significantly related to audit fees with a negative sign for all models. Auditors charge attractive fees during the first year of their mandate in order to retain customers. As expected, and in line with the results of André et al. (2016), December shows a positive sign, suggesting that the fees are higher when the audit is conducted during the busy season.

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5. ROBUSTNESS TESTS

5.1- The interaction between audit regulation and auditor industry expertise on audit fees

In order to examine the effects of the interaction between auditor quality (Big Four and auditor industry specialization) and the nature of audit regulation, I introduce in the initial model four variables (*R-Mandat*Auditor, Liability*Auditor*). The results are reported in Table 5. While *R-Mandat* is positively related to audit fees, the interaction between this variable and the quality of the auditor (*R-Mandat*Auditor*) does not influence the audit fees. I can therefore conclude that the possibility of annual renewal of the auditor's mandate is related to higher audit fees regardless of the type of auditor, Big or non-Big, specialist or non-specialist. This is not the case for the interaction between the auditors' civil liability (*Liability*) and type of auditor (*BIG4* and *SP-E*). This interaction (*Liability* * Auditor) is related to higher audit fees. In a country with the tort auditor's liability, Big Four firms and specialists at the European level (*SP-E*) charge higher audit fees.

[Insert Table 5]

5.2- Endogeneity

According to previous literature, empirical research is probably subject to two self-selection problems. The first is due to the fact that previous analysis covers all firms that have published audit fees; this publication should be mandatory or voluntary. The second problem concerns the endogeneity of auditor choice.

The obligation to publish the audit fees

To remedy this self-selection bias, I repeat all the previous regressions with a sample which excludes firms that did not have the obligation to publish the fees paid to their auditors (see Table 1. The results (not reported) are practically the same as those obtained previously with the entire sample. This comes from the low number of firms that voluntarily published the audit fees (700 cases out of 19,684 observations).

The choice of auditor and the fees of the audit

The second concern relates to the endogeneity of auditor choice (Choi et al. 2009; Johnstone 2014; Hsu et al. 2015). To control the possible effect of endogeneity, I use the two-step

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procedure of Heckman (1979). In the first step, I use the model *probit* explaining the auditor choice:

$$Prob[\text{BIG4 / SPEC} = 1] = Probit(\alpha_0 + \alpha_1 \text{ANTIDIR} + \alpha_2 \text{RLegal} + \alpha_3 \text{LnTA} + \alpha_4 \text{Lev} + \alpha_5 INREC + \alpha_6 \text{CFO_TA} + \alpha_7 \text{MTB} + \alpha_8 \text{Loss} + \alpha_9 \text{AbsAccT_TA} + \varepsilon)$$

BIG4 is 1 in the first model to explain the choice of a Big Four auditor; *SPEC* is 1 in a second model to explain the choice of a specialist. *ANTIDIR* is a dummy variable that is coded 1 if the protection investor index of LaPorta et al. (1998), revised by Pagano and Volpin (2005), is greater than 4. *Rlegal* is also a dichotomous variable. It is 1 if the index enforcement estimated by the average of three individual indices of La Porta et al. (1998) is higher than 9. *INREC* is the proportion of inventories and receivables in total asset. *Loss* is a dummy variable that takes the value 1 if the result is negative (Khurana and Raman 2004; Choi et al. 2008; Li 2009). Profitability is apprehended by the ratio of operating cash flow and total assets (*CFO_TA). MTB* is the market-to-book ratio. The absolute value of total *accruals* is normalized by total assets in the previous period (*AbsAccT_TA*).

Secondly, I introduce the Inverse Mills Ratio (IMR) generated from this regression model *probit* in models on the relationship between regulation and audit fees. Unreported results show that conclusions remain unchanged.

5.3- The effect of the size of the audited firm

The audited firm size is considered an important determinant of audit fees by previous literature (Hay et al. 2006). This importance is confirmed by determining the correlation matrix in which the correlation coefficient between the firm size (*LnTA*) and audit fees (*LnFees*) is 0.78. This correlation has a positive sign suggesting more the firm is large more audit fees are high. Given the importance of this factor, I empirically examined this relationship by dividing the sample into three sub-samples according to the size of the audited firm is small, medium or large.

[Insert Table 6]

The results of Table 6 confirm the importance of firm size in the determination of audit fees. Unlike auditor industry expertise (Big Four and specialists), which are positively and significantly related to audit fees for the three types of audited companies, the regulatory variables of the audit (except for *Joint-A*) do seem to play a role in the determination of audit fees for small firms, then to fully play to large firms. Tort liability (*Liability)* and possibility of annual renewal of the auditor's mandate (*R_Mandat)* are related to higher audit fees for large firms.

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5.4- The sample without UK

British firms represent 36.5% of the sample. To determine whether the results have been influenced by the dominance of British firms, I have re-estimated the regressions excluding these firms. The results (not reported) are similar to previous ones.

6. Conclusion

To the best of my knowledge, this study is the first to emphasize the importance of the respective impact of auditor characteristics and audit regulation on audit fees in Europe. Due to the analysis of international issues, this research offers a significant contribution to the academic literature dealing with audit fees. It draws on an extensive sample of European listed companies over a long period from 2003 to 2011. The main results show clearly that three regulatory variables—auditor liability, the annual renewal of the mandate, the joint audit—play a significant role in determining the audit fees. The possibility of annual renewal of the auditor's mandate involves higher fees. The fees are also higher when the auditor's liability is tort. To deal with potential disputes, auditors can enhance their account revision efforts or charge an additional premium to cover potential litigation risk (Choi et al. 2008; Ghosh and Pawlewicz 2009; André et al. 2016). Moreover, mandatory joint audit has a positive effect on audit fees. According to the results of Francis et al. (2005), Fung et al. (2012) and Carson (2009), the industry specialization auditors charge higher audit fees as well. The results are robust to several additional tests. The conclusions drawn from the analysis of the total sample remain persistent when replacing the study's original measure of auditor liability with an alternative measure, an index of liability of the auditor. They equally resist self-selection problems, namely the selection bias related to auditor choice and to the voluntary disclosure of audit fees.

This study has practical implications, as it may allow regulators to further explore the relationships between audit fees and auditor attributes and the different attributes of audit regulation, and also help the regulatory bodies to evaluate the coherence of regulations when revising them. This study may also offer insights for policy makers and market regulatory bodies interested in enhancing the audit market environment within Europe.

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Appendix: The variables of the study		
variables		Measure
Audit Fees	LnFees	Natural log of audit fees
Audit regulation attributes		
Tenure	R_Mandat	1 if maximum length of the audit mandate is 1 year, 0 otherwise
Auditor liability	Liability	1 if auditor liability is based on tort law, 0 otherwise
Joint Audit	Joint-A	Dummy variable coded 1 if joint audits were mandatory in year *t*, 0 otherwise.
The joint supply of audit and non-audit services	NAS	1 if the scope of non-audit services is restricted, 0 otherwise
Institutionnels Factors		
Gross Domestic Product	GDP	Annual metric variable taken from the World Bank database.
Investissement Direct à l'étranger	IDE	Net inflow (new investment inflows less disinvestment) of foreign investors in percentage of *GDP* in year *t*.
Auditor attributes		
Size of audit firm	BIG4	1 if the firm is audited by at least one Big auditor, 0 otherwise
European industry specialization of auditor	SP_E	1 if the auditor's market share in terms of total assets in firms in the same industry for the 15 EU countries under study≥ 10 %, 0 otherwise.
National industry specialization of auditor	SP_C	1 if the auditor's market share in terms of total assets of firms in the same industry and country ≥ 25%, 0 otherwise.
Audited firm attributes		
Firm size	LnTA	Logarithm of total assets
Auditee complexity	INREC	inventories +receivables / total assets
	SICRISK	1 if a firm's SIC code is 2833-2836, 3570-3577, 3600-3674, 5200-5961, 7370-7374 and 0 otherwise
	IFRS	1 if the firm adopted IFRS from 2005
Auditee risk	ROA	Earnings before extraordinary items deflated by prior period total assets
	LEV	Long term debt divided by total assets
	Loss	1 if the net result is negative; 0 otherwise.
Auditor-auditee relationship		
First year of engagement	P-Year	1 if this is the first year of the mandate;0 otherwise
Accounting closing date	December	1 if final year-end is 31 December; 0 otherwise

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Table 1- Value per country of the variables relating to audit regulation

	Mandatory audit firm tenure	Restriction non-audit services	Legalliability of auditor	Joint audit	mandatory *publication of audit fees*
	R_Mandat	**NAS**	**Liability**	**Joint-A**	**Fees_Pub**
Austria	0	0	0	0	1 since 2008
Belgium	1	1	1	0	1 since 2007
Denmark	0	0	0	0	1
Finland	0	0	1	0	1 since 2003
France	1	1	1	1	1 since 2003
Germany	0	0	1	0	1 since 2005
Greece	0	0	0	0	1 since 2009
Irland	0	1 since 2005	1	0	1
Italy	1	1	0	0	1 since 2006
Netherlands	0	0	0	0	1 since 2008
Portugal	1	0	1	0	1 since 2009
Spain	1	0	0	0	1 since 2003
Sweden	1	0	0	0	1
United Kingdom	0	1 since 2005	1	0	1

R_Mandat is coded 1 if the minimum duration of the audit mandate is mandatorily is 1 year, 0 otherwise. *NAS* is coded 1 if non-audit services are restricted, 0 otherwise. *Liability* is coded 1 if non-audit services are restricted, 0 otherwise. *Fees_Pub* is coded 1 if the publication of statutory audit fees and non-audit fees in the annual reports is mandatory, 0 otherwise.

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Table 2 -Descriptive statistics of variables (other than the attributes of audit regulation)

	Audit Fees (M€)	Total Assets (M€)	BIG4	SP-E	SP_C	INREC	SICRISK	IFRS	Perte	LEV	December	P-year
Austria	581,34	1 419 037,00	0,62	0,39	0,40	0,30	0,19	0,8	0,25	0,17	0,74	0,15
Belgium	12 769,59	1 545 042,00	0,50	0,31	0,27	0,32	0,16	0,69	0,32	0,16	0,86	0,14
Denmark	1 019,63	1 063 991,00	0,67	0,44	0,36	0,29	0,18	0,71	0,32	0,16	0,68	0,16
Finland	3 805,81	1 443 502,00	0,79	0,48	0,39	0,30	0,18	0,8	0,25	0,17	0,8	0,14
France	28 484,41	3 557 660,00	0,51	0,34	0,18	0,32	0,18	0,59	0,33	0,15	0,75	0,12
Germany	2 845,95	3 239 321,00	0,48	0,30	0,19	0,29	0,20	0,66	0,35	0,16	0,81	0,15
Greece	2 908,23	489 750,70	0,31	0,22	0,18	0,34	0,17	0,59	0,38	0,17	0,87	0,18
Irland	1 801,47	2 009 036,00	0,78	0,55	0,43	0,25	0,16	0,73	0,37	0,19	0,75	0,13
Italy	24 434,72	3 771 364,00	0,75	0,50	0,34	0,32	0,19	0,79	0,28	0,16	0,81	0,14
Netherlands	106 395,30	3 446 030,00	0,64	0,44	0,33	0,30	0,20	0,76	0,29	0,16	0,81	0,14
Portugal	1 267,99	2 598 058,00	0,57	0,42	0,48	0,29	0,11	0,76	0,28	0,21	0,84	0,16
Spain	10 992,56	5 831 777,00	0,72	0,45	0,41	0,28	0,16	0,76	0,20	0,16	0,77	0,12
Sweden	1 454,25	869 863,00	0,70	0,49	0,35	0,30	0,20	0,76	0,41	0,15	0,84	0,14
United Kingdom	2 926,42	1 593 132,00	0,54	0,33	0,22	0,27	0,17	0,70	0,42	0,16	0,62	0,13
Total	**9 711,57**	**2352521**	**0,56**	**0,36**	**0,25**	**0,29**	**0,18**	**0,69**	**0,36**	**0,16**	**0,74**	**0,14**

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Table 3- Pearson correlation matrix between variables

	LnFees	*BIG4*	*SP-E*	*SP-C*	*Liability*	*R_ Mandat*	*Joint-A*	*NAS*	*Rlegal*	*ANTIDIR*	*Ln-Taille*	*INREC*	*SICRISK*	*Loss*	*ROA*	*LEV*	*IFRS*	*P-year*	*December*
LnFees	1,000																		
BIG4	0,292	1,000																	
SP-E	0,241	0,621	1,000																
SP_C	0,268	0,357	0,368	1,000															
RLiability	-0,071	-0,086	-0,074	-0,12	1,000														
R-Mandat	-0,112	-0,031	-0,034	-0,04	0,384	1,000													
Joint-A	0,115	-0,05	-0,032	-0,08	0,248	-0,555	1,000												
NAS	-0,148	-0,068	0,006	-0,088	0,34	0,454	-0,445	1,000											
Rlegal	-0,119	-0,025	-0,027	0,001	0,208	0,739	-0,643	0,377	1,000										
ANTIDIR	-0,117	0,013	-0,034	-0,02	0,294	0,447	-0,362	0,367	0,275	1,000									
LnTaille	0,787	0,286	0,239	0,316	-0,14	-0,171	0,065	-0,127	-0,179	-0,17	1,000								
INREC	-0,11	-0,015	-0,027	-0,044	-0,022	-0,024	0,02	-0,037	-0,008	-0,072	-0,147	1,000							
SICRISK	-0,016	-0,014	-0,017	0,003	-0,001	0,007	0,003	0,012	0,008	-0,005	-0,037	0,064	1,000						
Perte	-0,178	-0,072	-0,064	-0,073	0,042	0,047	-0,022	0,038	0,044	0,051	-0,253	-0,063	0,032	1,000					
ROA	0,16	0,076	0,061	0,066	-0,044	-0,049	0,023	-0,042	-0,048	-0,053	0,242	0,07	-0,038	-0,661	1,000				
LEV	0,17	0,055	0,065	0,075	-0,026	-0,033	0,009	-0,025	-0,039	-0,031	0,205	-0,208	-0,082	-0,019	-0,008	1,000			
IFRS	0,016	-0,017	0,123	-0,034	-0,123	-0,247	0,145	0,055	-0,219	0,206	-0,018	-0,04	-0,004	0,009	-0,01	0,013	1,000		
P-year	-0,028	-0,049	-0,049	-0,02	-0,004	0,008	-0,004	0,007	0,006	-0,012	-0,016	-0,004	0,001	0,029	-0,031	-0,006	0,01	1,000	
December	0,091	0,018	0,012	0,037	-0,062	-0,064	0,024	-0,042	-0,063	-0,113	0,1	-0,009	-0,003	-0,001	0,011	0,026	-0,014	0,03	1,000

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Table 4- Impact of audit regulation on audit fees

Dependent variable ***LnFees***

	Auditor Liability			***Tenure***			***Joint Audit***			***Joint Effect of audit regulation attributes***		
	BIG4	SP-E	SP_C	BIG4	SP-E	SP_C	BIG4	SP-E	SP_C	***BIG4***	SP-E	SP_C
Intercept	-2,0733	-2,0309	-2,0589	-1,9744	-1,9288	-1,9742	-1,701	-1,6518	-1,6874	-2,3971	-2,3644	-2,3879
	(-25,94)***	(-25,14)***	(-25,26)***	(-23,20)***	(-22,67)***	(-23,83)***	(-22,29)***	(-20,91)***	(-21,32)***	(-24,60)***	(-23,75)***	(-23,87)***
BIG4/SPEC	0,2809	0,1941	0,0976	0,0782	0,1831	0,2681	0,2891	0,2042	0,1039	0,2825	0,1858	0,095
	(14,03)***	(8,54)***	(4,04)***	(3,27)***	(8,05)***	(13,55)***	(14,17)***	(8,69)***	(4,32)***	(14,43)***	(8,34)***	(3,93)***
Liability	0,198	0,1885	0,1915							0,1517	0,1415	0,1422
	(6,94)***	(6,59)***	(6,79)***							(5,14)***	(4,82)***	(4,89)***
R_Mandat				0,1318	0,1247	0,1256				0,1931	0,1866	0,1988
				(5,01)***	(4,73)***	(4,81)***				(6,38)***	(6,19)***	(6,58)***
Joint-A							0,3592	0,3586	0,3485			
							(10,42)***	(10,32)***	(10,23)***			
NAS				-0,0302	-0,0361	-0,0164	-0,0777	-0,0983	-0,0868			
				(-1,36)	(-1,66)	(-0,75)	(-3,58)***	(-4,55)***	(-3,93)***			
LnTA	0,6068	0,6134	0,6174	0,6169	0,6119	0,6055	0,5948	0,6013	0,6056	0,6048	0,6123	0,616
	(119,96)***	(124,06)***	(124,24)***	(118,96)***	(117,38)***	(114,60)***	(114,79)***	(116,39)***	(117,90)***	(115,71)***	(118,88)***	(119,15)***
INREC	0,1148	0,1251	0,1331	0,1254	0,1163	0,1065	0,0579	0,0684	0,0784	0,1168	0,1288	0,1364
	(1,72)*	(1,86)*	(2,01)**	(1,93)*	(1,76)*	-1,63	-0,86	-1,01	-1,17	(1,76)*	(1,92)**	(2,06)**
SICRISK	0,063	0,0629	0,0589	0,0548	0,0582	0,0582	0,0586	0,0585	0,0548	0,0644	0,0639	0,0602
	(2,17)**	(2,18)**	(2,04)**	(1,91)*	(2,03)**	(2,01)**	(1,97)*	(2,00)**	(1,87)*	(2,23)**	(2,23)**	(2,10)**
Loss	0,0902	0,091	0,0891	0,0899	0,0917	0,091	0,101	0,1019	0,0996	0,093	0,0939	0,0918
	(3,32)***	(3,36)***	(3,23)***	(3,31)***	(3,43)***	(3,40)***	(3,65)***	(3,70)***	(3,56)***	(3,48)***	(3,52)***	(3,39)***
Lev	0,0151	0,0141	0,0148	0,0153	0,0145	0,0155	0,0156	0,0145	0,0153	0,0155	0,0146	0,0153
	(2,37)**	(2,25)**	(2,37)**	(2,49)**	(2,35)**	(2,46)**	(2,43)**	(2,30)**	(2,45)**	(2,38)**	(2,28)**	(2,38)**
IFRS	0,1519	0,1206	0,1516	0,1683	0,1388	0,1641	0,0906	0,0625	0,0939	0,3131	0,2728	0,3116
	(5,63)***	(4,38)***	(5,56)***	(5,61)***	(4,61)***	(5,58)***	(3,30)***	(2,23)**	(3,36)***	(7,96)***	(7,01)***	(7,91)***
P_Year	-0,0645	-0,0681	-0,0768	-0,0783	-0,07	-0,0666	-0,0648	-0,068	-0,0769	-0,0607	-0,0648	-0,0732
	(-2,35)**	(-2,47)**	(-2,74)***	(-2,76)***	(-2,51)**	(-2,39)**	(-2,42)**	(-2,52)***	(-2,80)***	(-2,24)**	(-2,38)**	(-2,64)***
December	0,0663	0,0651	0,0625	0,0585	0,0605	0,0613	0,0503	0,0498	0,0475	0,0708	0,0698	0,0673
	(2,66)***	(2,66)***	(2,56)**	(2,39)**	(2,47)**	(2,48)**	(2,05)**	(2,05)**	(1,97)*	(2,82)***	(2,83)***	(2,73)***
GPD	0,0069	0,0068	0,0071	0,0074	0,0071	0,0072	0,0097	0,0094	0,0096	0,0051	0,0051	0,0053
	-0,94	-0,99	-1,07	-1,16	-1,08	-1,04	-1,29	-1,33	-1,41	-0,68	-0,72	-0,77
IDE	-0,0018	-0,0017	-0,0018	-0,0016	-0,0014	-0,0016	-0,0015	-0,0013	-0,0015	-0,0014	-0,0013	-0,0014
	(-2,15)**	(-1,96)*	(-2,25)**	(-1,99)**	(-1,73)*	(-1,93)*	(-1,78)*	(-1,57)	(-1,88)*	(-1,73)*	(-1,56)	(-1,83)*
Fixedeffects	years	years	years	years	years	years	years	years	years	years	years	years
R2	0 .6281	0.6257	0.6236	0.6235	0.6255	0.6278	0.6303	0.6279	0.6256	0.6324	0.6297	0.6278
N	11372	11372	11372	11433	11433	11433	11515	11515	11515	11372	11372	11372

***, ** and * indicate statistical significance respectively at the 1%, 5% and 10%.

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Table 5- Interaction of audit regulation and auditor quality on the audit fees

Dependent variable ***LnFees***

	BIG4	***SP-E***	***SP_C***
Intercept	-2,4519	-2,3794	-2,4158
	(-16,03)***	(-18,91)***	(-19,72)***
BIG4/SPEC	0,4175	0,2535	0,1600
	(2,46)**	(1,61)	(0,99)
Liability	-0,0371	0,0642	0,1254
	(-0,71)	(1,68)*	(3,76)***
R_Mandat	0,2033	0,1994	0,1794
	(4,16)***	(5,94)***	(4,37)***
Liability*Auditor	0,2770	0,1618	0,0522
	(4,65)***	(2,86)***	(0,89)
R_Mandat*Auditor	-0,0307	-0,0270	0,0431
	(-0,46)	(-0,52)	(0,70)
Cotrol Variables	Yes	yes	yes
Fixed effects	Years	years	years
R2	0.6336	0.6303	0.6283
N	11372	11372	11372

***, ** and * indicate statistical significance respectively at the 1%, 5% and 10%.

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Table 6- Impact of auditee firms size on audit fees

Dependent variable ***LnFees***

	Small-Sized Firms						***Mid-Sized Firms***						**Large firms**					
	BIG4		***SP_E***		***SP_C***		***BIG4***		***SP_E***		***SP_C***		***BIG4***		***SP_E***		***SP_C***	
Intercept	0,2549	0,5166	0,3393	0,5861	0,2704	0,5616	-1,5329	-1,8601	-1,5463	-1,8839	-1,6164	-1,9575	-2,8219	-3,7062	-2,6756	-3,5937	-2,7520	-3,6637
	(1.00)	(1.19)	(1.31)*	(1.35)	(1.02)	(1.29)	(-3.94)***	(-4.64)***	(-3.92)***	(-4.64)***	(-4.16)***	(-4.90)***	(-16.67)***	(-20.64)***	(-15.67)***	(-19.76)***	(-15.94)***	(-19.85)***
Auditor	0,4917	0,4542	0,4071	0,3789	0,2709	0,2891	0,3046	0,3083	0,2016	0,2004	0,1368	0,1332	0,2476	0,2247	0,1641	0,1289	0,0062	-0,0134
	(11.10)***	(10.44)***	(7.16)***	(7.14)***	(3.93)***	(3.99)***	(9.77)***	(10.26)***	(5.86)***	(5.81)***	(4.48)***	(4.38)***	(7.63)***	(6.75)***	(5.43)***	(4.50)***	(0.17)	(-0.38)
Joint-A	0,3141		0,3411		0,3480		0,1235		0,1215		0,1178		0,4971		0,4905		0,4644	
	(4.61)***		(4.86)***		(4.75)***		(2.17)**		(2.14)**		(2.08)**		(11.14)***		(10.87)***		(10.63)***	
NAS	-0,2401		-0,2926		-0,3087		-0,0303		-0,0637		-0,0590		-0,0885		-0,0946		-0,0829	
	(-2.49)**		(-3.00)***		(-3.15)***		(-0.88)		(-1.84)*		(-1.70)*		(-2.48)**		(-2.67)***		(-2.32)**	
Liability		-0,0720		-0,1060		-0,1349		0,0277		0,0099		0,0099		0,2097		0,2091		0,2060
		(-0.64)		(-0.96)		(-1.22)		(0.57)		(0.20)		(0.20)		(5.34)***		(5.38)***		(5.32)***
R-Mandat		-0,0673		-0,1030		-0,0575		0,1771		0,1725		0,1859		0,2149		0,2149		0,2275
		(-0.54)		(-0.83)		(-0.46)		(3.89)***		(3.81)***		(4.08)***		(5.70)***		(5.72)***		(6.19)***
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	years	years	years	years	years	years	Years	years	years	years	years	years	years	years	years	years	years	years
R2	0.2494	0.2535	0.2189	0.2356	0.2024	0.2233	0.1215	0.1236	0.1108	0.1122	0.1056	0.1070	0.4602	0.4653	0.4584	0.4634	0.4559	0.4618
N	1791	1772	1805	1772	1805	1772	3843	3812	3843	3812	3843	3812	5867	5788	5867	5788	5867	5788

***, ** and * indicate statistical significance respectively at the 1%, 5% and 10%.

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